Python Lab: Input, Output, Comments & Data Types

This lab covers fundamental Python concepts, including input/output operations, comments, type casting, and basic calculations. Complete the exercises according to your comfort level but challenge yourself to try the next level!

Learning Objectives

By completing this lab, you will:

- Understand how to use `input()` to get user data
- Learn to use `print()` for output
- Master type casting with `int()`, `float()`, and `str()`
- Practice writing clear code comments
- Develop problem-solving skills appropriate to your level

BEGINNER LEVEL

Perfect if you're new to programming or Python.

Exercise 1: Personal Greeting

Write a program that:

- Ask the user for their first name
- Asks for their last name
- Prints: "Hello, [FirstName] [LastName]! Welcome to Python programming."

Sample Output:

```
Hello, Sarah Johnson! Welcome to Python programming.
```

Exercise 2: Favorite Things

Create a program that:

- Ask the user for their favorite color
- Asks for their favorite food
- Prints both answers in a friendly sentence

Sample Output:

Your favorite color is blue, and you love eating pizza!

Exercise 3: Simple Calculator

Write a program that:

- Includes a comment at the top explaining what the program does
- Asks the user for two numbers
- Converts them into integers
- Prints their sum

Sample Input and Output:

Enter first number: 10

Enter second number: 25

The sum is: 35

Exercise 4: Age Calculator

Write a program that:

- Asks the user for their current age
- Converts it to an integer
- Calculates their age in 10 years
- Prints: "In 10 years, you will be [age] years old."

INTERMEDIATE LEVEL

For students with some basic programming knowledge.

Exercise 1: Student Information System

Write a program that:

- Asks for the student's name, student ID, and program of study
- Asks for their age and converts it to an integer
- Calculates their birth year (current year is 2025)
- Prints a formatted summary with all the information

Sample Input and Output:

```
Enter Student Name: Ahmad bin Ali
Enter Student ID: 2025001234
Enter Student Program: Computer Science
Enter Student Current Age: 19

=== Student Profile ===
Name: Ahmad bin Ali
Student ID: 2025001234
Program: Computer Science
Age: 19
Birth Year: 2006
===================
```

Exercise 2: Multi-Operation Calculator

Create a program with comments that:

- Asks the user for two decimal numbers
- Converts them to floats
- Calculates them and displays:
    - Addition
    - Subtraction
    - Multiplication
    - Division

Sample Input and Output:

Enter first number: 12.5

Enter second number: 4.2

Addition: 16.7

Subtraction: 8.3

Multiplication: 52.5

Division: 2.976190476190476

Exercise 3: Shopping Calculator

Write a program that:

- Asks for the price of an item (as a float)
- Asks for the quantity (as an integer)
- Calculates the subtotal
- Calculating a 6% sales tax
- Calculates and displays the total amount

Sample Input and Output:

Item price: 29.99

Quantity: 3

Subtotal: RM 89.97

Tax (6%): RM 5.40

Total: RM 95.37

Exercise 4: Temperature Converter

Create a program that:

- Includes clear comments explaining each section
- Ask the user for the temperature in Celsius
- Convert it into a float
- Convert it to Fahrenheit using the formula: $F = (C \times 9/5) + 32$
- Convert it to Kelvin using the formula: $K = C + 273.15$
- Displays both conversions with 2 decimal places

ADVANCE LEVEL

Challenge yourself with more complex calculations and formatting.

Exercise 1: Student Performance Report

Write a well-commented program that:

- Asks for the student's name and TP number
- Asks for marks for 5 subjects (convert to float)
- Calculates the total marks
- Calculates the average marks
- Calculates the percentage (out of 500 total marks)
- Displays a formatted report card with all information

Sample Output:

```
=== STUDENT PERFORMANCE REPORT ===
Student Name: Nurul Aina
TP Number: TP098776
========================================
Mathematics: 85.5
Physics: 78.0
Chemistry: 92.0
Programming: 88.5
English: 76.0
========================================
Total Marks: 420.0 / 500
Average: 84.0
Percentage: 84.0%
========================================
```

Exercise 2: Comprehensive Loan Calculator

Create a program that:

- It has comprehensive comments explaining the loan calculation
- Asks for loan amount (principal) as a float
- Asks for annual interest rate (as a percentage) as a float
- Asks for loan period in years as an integer
- Calculates the following:
  - Total interest = Principal × (Rate/100) × Years
  - Total amount to repay = Principal + Total interest
  - Monthly payment = Total amount / (Years × 12)
  - Daily saving needed = Total amount / (Years × 365)
- Displays all calculations in a well-formatted output with RM currency

Sample Input and Output:

Loan Amount: <u>50000.00</u>

Interest Rate: <u>5.5% per year</u>

Loan Period: <u>3</u>

```
=== LOAN CALCULATION SUMMARY ===
Loan Amount: RM 50000.00
Interest Rate: 5.5% per year
Loan Period: 3 years
--------------------------------------------------------------
Total Interest: RM 8250.00
Total Repayment: RM 58250.00
Monthly Payment: RM 1618.06
Daily Saving Needed: RM 53.20
========================================
```

Exercise 3: Advanced Time Converter

Write a program that:

- Ask the user to enter seconds (as an integer)
- Convert seconds into:
  - Weeks
  - Days (remaining after weeks)
  - Hours (remaining after days)
  - Minutes (remaining after hours)
  - Seconds (remaining after minutes)
- Also calculating the total in other single units (total minutes, total hours, total days)
- Displays the complete breakdown

Hint: Use integer floor division (//) and modulo (%) operators

Sample Input and Output:

```
Enter total seconds: 1000000
=== TIME BREAKDOWN ===
1000000 seconds equals:
Weeks: 1
Days: 4
Hours: 13
Minutes: 46
Seconds: 40
=== ALTERNATIVE REPRESENTATIONS ===
Total Minutes: 16666.67
Total Hours: 277.78
Total Days: 11.57
========================================
```

Tips for Success

- **Beginners:** Start with the beginner level and make sure you understand each concept before moving on
- **Intermediate:** Try to complete beginner exercises first for practice, then move to intermediate
- **Advanced:** Challenge yourself with advanced exercises, but review the basics if needed
- **Everyone:** Add comments to explain your code - this helps you and others understand your logic!